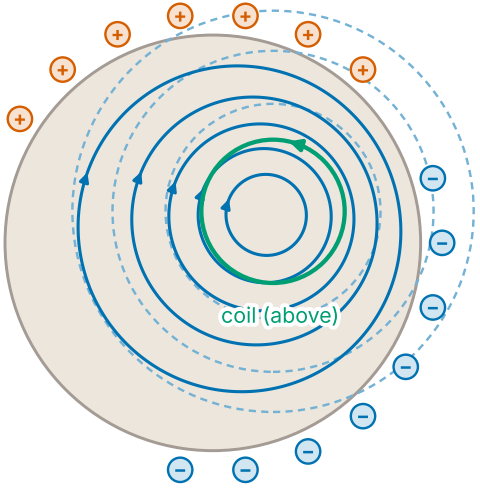


primary field $\mathbf{E}_p = -\partial_t \mathbf{A}$

set by the coil alone, circling its axis;
it does not know where the conductor ends.

boundary charge

no current can leave the rim; charge piles up:
positive where $\mathbf{E}_p$ points out, negative where in.

total field $\mathbf{E} = \mathbf{E}_p + \mathbf{E}_s$

the charges' secondary field $\mathbf{E}_s$ bends E to run along the rim; stimulation and heating see this field.

charge feeds back on E

$I(t)$ coil current → $\mathbf{B}(t)$ magnetic field → $\partial_t \mathbf{B} \to \mathbf{E}_p$ induced field → $\mathbf{J} = \sigma \mathbf{E}$ tissue current → $q_s \to \mathbf{E}_s$ boundary charge → $\mathbf{E} = \mathbf{E}_p + \mathbf{E}_s$ nerves, heat